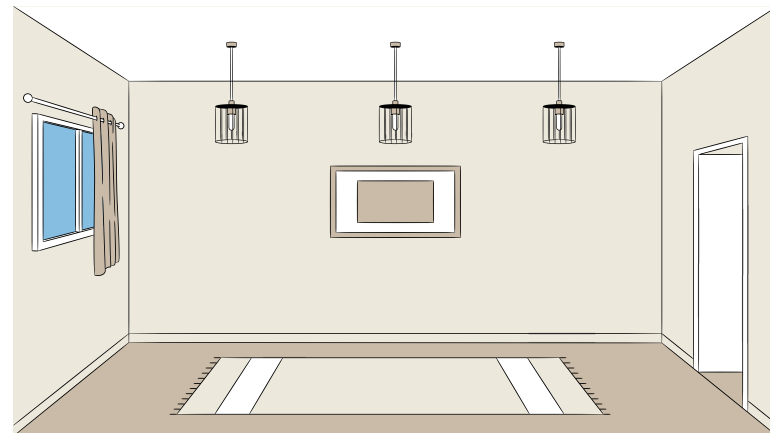


The 3 Musketeers of Web Dev

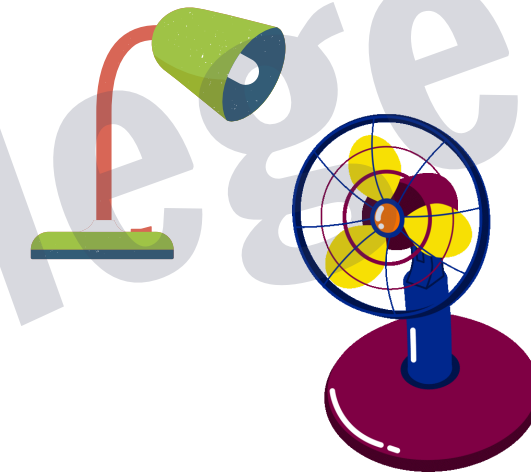
HTML
(structure)



CSS
(style)



JS
(logic)



Starter Code

<style> tag connects HTML with CSS

<script> tag connects HTML with JS

Apna College

```
<html>
```

```
  <head>
```

```
    <title> Website Name </title>
```

```
  </head>
```

```
  <body>
```

```
    <!-- Content Tags -->
```

```
  </body>
```

```
</html>
```

you can write js code like prompt or loops strings or alert etc inside the script tag only , if u dont want to make a new file and link it to the html file

like

```
<body>  
<script>  
alert("hello");  
</script>  
</body>  
</html>
```

Window Object

The window object represents an open window in a browser. It is browser's object (not JavaScript's) & is automatically created by browser.

(can be asked in interviews so remember these 2 lines)

It is a **global** object with lots of properties & methods.

it is a browsers object (the window is a object name present in browser or in any web page)
go type window in the console ,,
itll show a object with alot of functions in it

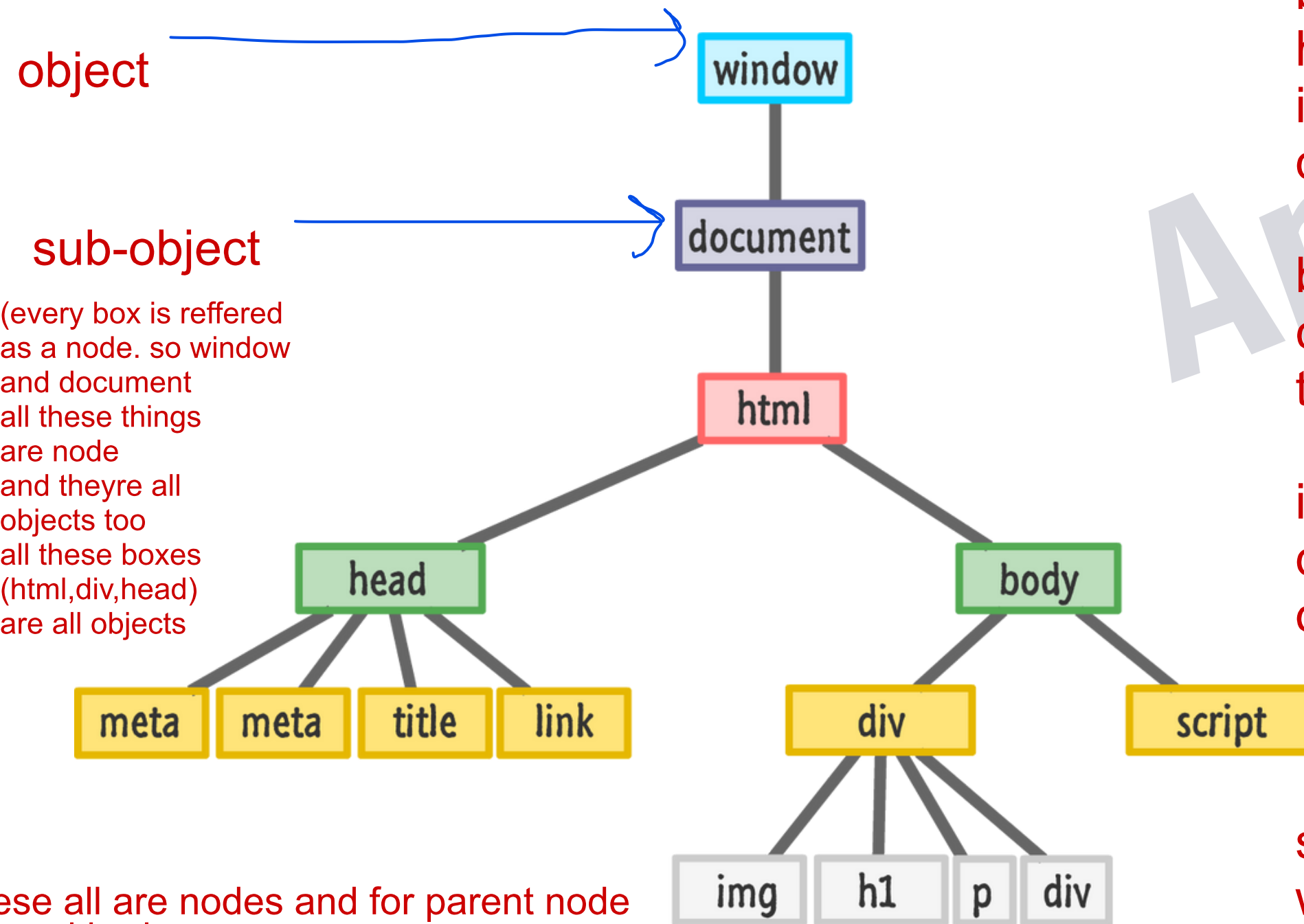
"by browser" so if using chrome , chrome makes it and if u using edge edge makes it (makes window object);

every code you write in script.js or any js code ,, all that code comes under window object which is automatically created , so all the js code you right is inside and is a part of window object see code line 1 and 2 theyre the same. you dont need to write window. coz the code automatically understands whatever youre writing is inside the window object only.

this tree like structure is exactly what the DOM is , document object model is a tree like structure shown below

What is DOM?

When a web page is loaded, the browser creates a **Document Object Model (DOM)** of the page



basically we all know that html css js are all part of the html code only.....
if you want to get the html code from the js file you can do it...

because as yk window is the largest object it has all the code inside it html code js code .. all the code is inside the window object but where in it?

inside the window object, there's another object named document , so document is an object inside the window object which stores the html code, only html code it stores

so the Document Object Model is:
window-->document-->html code

there's also a code as `console.dir` unlike `console.log` which is used to print all the properties and methods of some special objects like here for ex document you can do `console.dir(window.document)` to see the properties and methods of the document object.

these all are nodes and for parent node named body, div and script are the child nodes and div and script are siblingNodes to each other

DOM Manipulation

Selecting with id

```
document.getElementById("myId")
```

Selecting with class

```
document.getElementsByClassName("myClass")
```

Selecting with tag

```
document.getElementsByTagName("p")
```

how will the query selector identify if its a tag or a class or a id?

by the syntax of class and id , for tag its just the tags name for but class u write .class and id #id

DOM Manipulation

Query Selector

by writing `document.querySelector("");`
inside you can write anything from 3
like `tag` `className` or `idName`. 3 in 1
but for class and tag name it won't return all
the elements just the first tag for tag and first
element for class

`document.querySelector("#myId / .myClass / tag")`

//returns first element

`document.querySelectorAll("#myId / .myClass / tag")`

//returns a NodeList

→ meaning, it returns all the elements

for id you can use any of these both are same but for class and tags they differ

DOM Manipulation

you can also get these property values and also change these values

Properties

3 types of nodes namely, text, comment and element we only deal with element node.

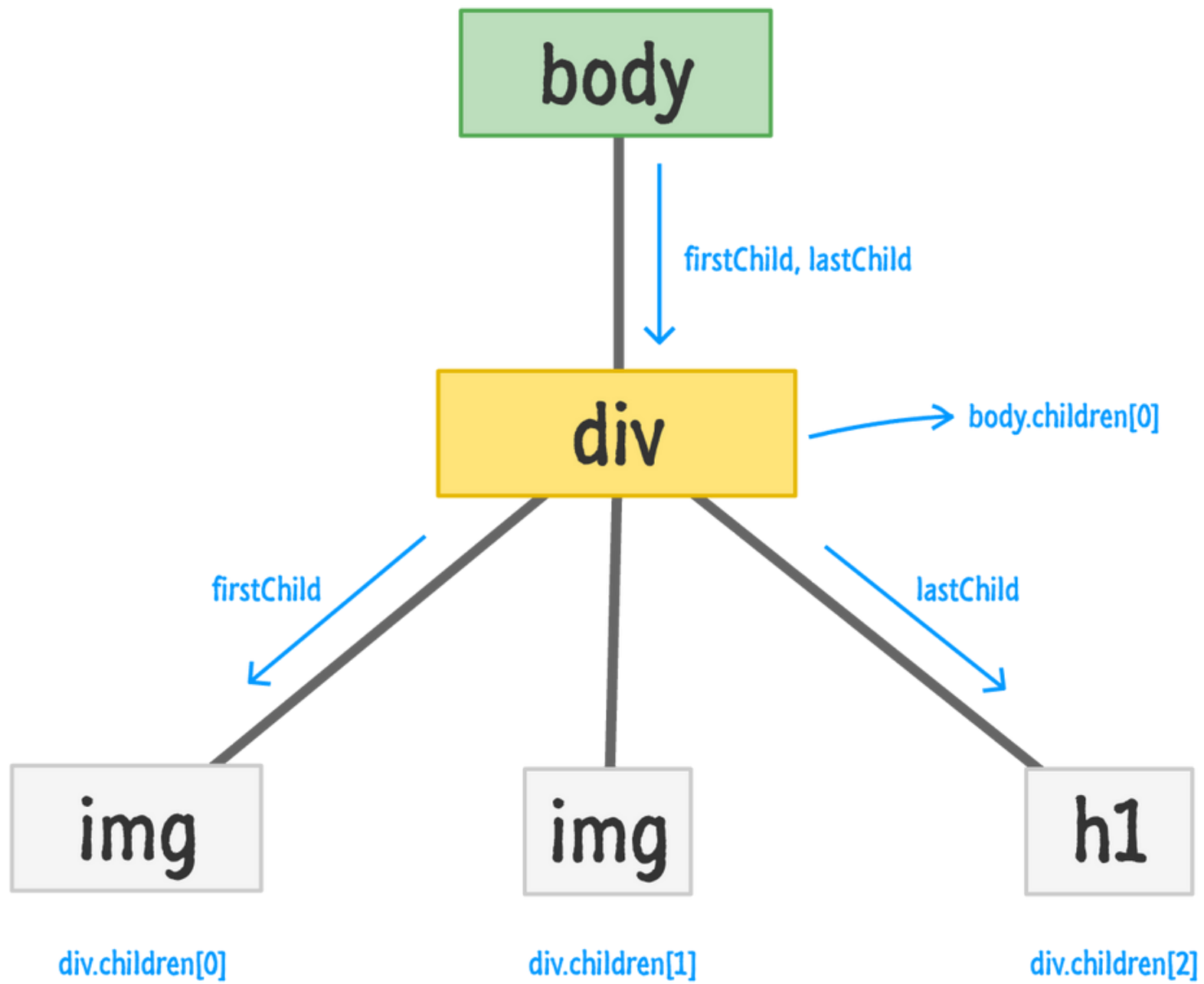
- **tagName** : returns tag for element nodes (only for tag, id but not for class)
- **innerText** : returns the text content of the element and all its children (only texts not even children tags showing only their texts
codeline:131)
- **innerHTML** : returns the plain text or HTML contents in the element returns all the code like all the children tags too but not parent tag
- **textContent** : returns textual content even for hidden elements

innerText and textContent me kya fark hai?

textContent innerText jesa hi hai bas hidden elements keliye bhi hai

for style="visibility:hidden" tags textContent shows the text and innerText doesn't but innerHTML does, so innerHTML and textContent both show the text so for only seeing hidden elements' text do textContent and for seeing all the tags too do innerHTML

Homework



Let's Practice

Qs. Create a H2 heading element with text - "Hello JavaScript". Append "from Apna College students" to this text using JS.

append means add from last

q1 very imp pls see

concatenate elements text.

Qs. Create 3 divs with common class name - "box". Access them & add some unique text to each of them.

 `<img src="">`
here src is attribute

DOM Manipulation

In HTML, the part of a tag that provides additional information about an element is called an attribute like `id="heading"` `<p yahape kuchbhi is attribute></p>`

Attributes

- `getAttribute("attr")` //to get the attribute value
- `setAttribute("attr","value")` //to set the attribute value

`node.getAttribute(attr)`
`node.setAttribute(attr,value)`
node here is the tags name

Style

- `node.style` // to get the css styling of that tag ,,,, `tag.style`

```
let tag=document.querySelector(".hheading");
```

// works only for inline styling , means style attribute ka use krke jo styling ki gayi hai sirf uske liye hi ye node.style work karega

you can also change the inline styling from js file by this node.style
how?

```
type node.style.color="red";
```

```
node.style.backgroundColor="red";
```

note its not background-color but backgroundColor

for css properties and their names in js heres how it goes

DOM Manipulation

Attributes

- `getAttribute( attr )` //to get the attribute value
- `setAttribute( attr, value )` //to set the attribute val th

Style

- `node.style`

color → color
background-color → backgroundColor
font-size CSS → font Size
js



note: insert an element and remove are the only property jisme double apostrophe nahi aata hai andar

DOM Manipulation

means(xyz); not ("xyz")

insert node/tag

inserting is a 2step process
where you first create then insert
1.create
2.insert

Insert Elements

let el = document.createElement("div")

- `node.append( el )` //adds at the end of node (inside) (line code 235 se 244) (also check the html code comments)
- `node.prepend( el )` //adds at the start of node (inside) (line code 246 se 250)
- `node.before( el )` //adds before the node (outside)
- `node.after( el )` //adds after the node (outside)

Delete Element

(delete node/tag)

- `node.remove( )` //removes the node

tag.remove(); thats it

let tag=document.querySelector("p");

Let's Practice

H.w

appendChild();
removeChild();

study about these elements from MDN documentation

Qs. Create a new button element. Give it a text “click me”, background color of red & text color of white.

Insert the button as the first element inside the body tag.

Qs. Create a <p> tag in html, give it a class & some styling.

means adding to <p >at the end meaning the existing class to be there and also at the end adding another class. not node.append but adding it as a attribute in <p ...></p>

Now create a new class in CSS and try to append this class to the <p> element.

Did you notice, how you overwrite the class name when you add a new one?

by setAttribute

Solve this problem using classList.

to add a class more or remove...

```
node.classList.add("xyz");  
node.classList.remove("xyz");
```

see the code style2.css